



The Scope of Electronic Vehicles in Nepal

Pravakar Bogati*, Prasiddha Raut, Pratyoosh Dahal

Department of Mechanical Engineering, School of Engineering, Kathmandu University, Nepal

With Nepal, with its population surpassing 30 million and its growing economy, faces escalating environmental challenges due to the increasing use of fossil-fueled vehicles. Recognizing the urgency, the government is actively promoting electric cars (EVs) as a long-term solution. This comprehensive study delves into Nepal's EV potential, shedding light on its transportation landscape and the obstacles it encounters. The research employs a multifaceted methodology, including a Consumer Preference Survey to gauge public sentiment towards EVs. This involves assessing acceptance levels, addressing concerns, and capturing expectations regarding EV adoption. An Energy Cost Analysis compares gasoline, diesel, and electricity costs nationwide to evaluate the economic feasibility of EVs compared to conventional vehicles. Import Regulations and Policy are scrutinized to understand the regulatory environment influencing EV accessibility and affordability, encompassing incentives, tariffs, and trade policies. Additionally, a Driving Preferences Study investigates whether the populace leans towards the eco-friendly appeal of EVs or prefers traditional combustion engine vehicles. The findings are rigorously analyzed through data modeling software. Notably, the study identifies key impediments to EV adoption: the scarcity of charging stations and limited consumer understanding of EV product quality. This research provides invaluable insights for policymakers, stakeholders, and the general public. It underscores the necessity of collaborative efforts involving governments, researchers, consumers, and stakeholders to realize the benefits of electric vehicles, address the associated challenges, and expedite the transition towards sustainable and clean transportation in Nepal. It also sparks crucial discussions about Nepal's electric vehicle future.

Introduction

Nepal, with 30+ million people and 1.7% annual growth, sees rising per capita income (\$1,336.5). Growing economy impacts vehicle ownership, favoring fossil-fueled cars due to increased petroleum imports. Two-wheelers own 27.3% of the market, while three/four-wheelers represent 3.1%. Fossil fueled foreign vehicle imports dominate, causing environmental pollution. Nepal targets carbon neutrality by 2045; EVs could reduce 37.8% of 2016 CO2 emissions. In 2021, electricity usage was 4.2%, transport heavily reliant on petroleum (9%). Despite potential for 103 GW electricity, current capacity is 2,191 MW, mostly from hydropower. The introduction of EVs could bridge this gap, with the study addressing EV status, public perspectives, price comparisons, charging station penetration, and government regulations.

Methodology

A mixed-method approach was employed to gather insights into customer perspectives and behaviors regarding Electric Vehicles (EVs) in Nepal. This comprehensive methodology included a survey using online questionnaires and email links, ensuring anonymity and broad reach. A robust theoretical framework synthesized prior research to guide the study and identify key issues. Additionally, a comprehensive case study on Nepal explored its EV history, current market dynamics, government policies, and barriers to green transportation. The research design involved an online questionnaire distributed through social media, focusing on demographic data and questions about vehicle engagement, mobility patterns, EV perspectives, adoption predictions, investment willingness, and barriers to EV infrastructure development. Data was analyzed to understand consumer behavior and support sustainable development in emerging economies, offering a multi-faceted approach to investigate EV perceptions and behaviors in Nepal.

Description

Challenges include a lack of charging infrastructure, limited public awareness, and higher initial purchase costs. Policy improvements and awareness campaigns are vital for sustainable growth. Here is a Table showcasing important policies driving the EV Market.

Table 1.1 Important Policies driving the EV Market

Policy/Initiative	Description
Net-Zero Carbon Emission Commitment (COP26)	Nepal committed to achieving net-zero carbon emissions by 2045 at the COP26 in Glasgow (November 2021).
Nationally Determined Contribution (NDC)	Nepal's NDC includes targets to increase EV adoption and reduce reliance on fossil fuels by 50%.1
EV Adoption Targets	The government aims for 25% EV usage by 2025 and 90% by 2030.
Recent Import Trends	Remarkable growth in the EV market, with NPR 11.23 billion worth of EVs imported in the first 11 months of FY 2022/23.
Excise and Customs Waiver (FY 2023/24)	Waiver on excise and customs for EVs up to 100 kW, with varying excise duties for higher-capacity EVs.
Fast Charging Stations	Nepal Electricity Authority (NEA) inaugurated 51 fast charging stations with concessional loans and technical support.

Results

Result Shows that the public interests in EV primarily. A 95% Confidence interval is subjected to the survey .

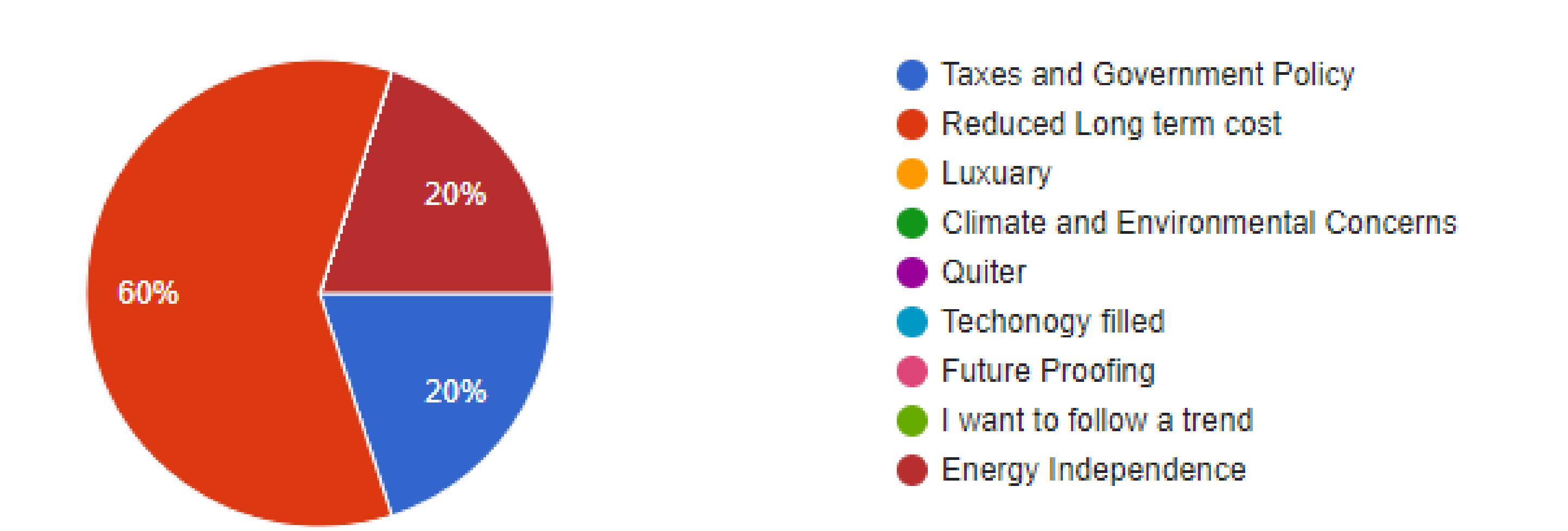


Figure 1.1 Survey shows factor consumers would look before buying an EV.



Figure 1.2 Increasing daily transactions from EV Stations across Nepal

Conclusions

Findings: Nepal's growing population and economy favor fossil-fueled cars while the younger generations favor two-wheelers EVs. EVs can cut emissions and reduce reliance on fuel imports. Insufficient charging infrastructure, low public awareness, and high initial costs are the major challenges. Government commitment to carbon neutrality by 2045 and pro-EV policies show promise. Nepal's electricity generation capacity lags behind potential, making EVs a viable energy solution.

Future Outcomes:

Policies and incentives must make EVs more appealing. Awareness campaigns are vital. Renewable energy integration can drive sustainable EV growth. Collaboration among stakeholders is essential for Nepal's EV success.

References

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